

THURSDAY • JUNE

Seasonal VariationsMeasurement of Seasonal variations.

Most of the phenomena in economics and business show seasonal patterns, when data are expressed annually there is no seasonal variation.

However, monthly or quarterly data frequently exhibit strong seasonal movements and considerable interest attaches to devising a pattern of average seasonal variation.

Eg:- If we observe the sales of a bookseller we find that for the quarter July-September (when most of the students purchase books), sales are maximum. If we know by how much the sales of this quarter are usually above or below the previous quarter for seasonal reasons, we shall be able to answer a very basic question, namely, was this due to an underlying upward tendency or simply because this quarter is usually seasonally higher than the previous quarter.

Before attempting to measure seasonal variation certain preliminary

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decisions must be made. For example, It is necessary to decide whether weekly, quarterly or monthly indexes are required. This will be decided in the light of the nature of the problem and the type of data available. Seasonal adjustments help avoid misinterpretation. →

To obtain a stable description of a pattern of seasonal variation it will be desirable to first free the data from the effects of trend, cycles and irregular variation. Once these other components have been eliminated, we can calculate, in index form, a measure of seasonal variations which is usually referred to as a Seasonal index. Thus the measures of seasonal variations are called Seasonal indexes (per cent).

⇒ A Specific seasonal index refers to the seasonal changes during a particular year.

⇒ A typical Seasonal index is obtained by averaging a number of specific Seasonals. It is thus a generalized expression of seasonal variations for a series.

The following are some of the methods more popularly used for measuring seasonal variations :-

1) Method of Simple averages (weekly, monthly or quarterly)

2) Ratio - to - trend method

3) Ratio - to - Moving Average method

4) Link Relative method

Method of Simple averages

The following ^{steps} are necessary for calculating the index:

1) Arrange the unadjusted data by years and months (or quarters if quarterly data are given).

2) Find the totals of January, February etc.

3) Divide each total by the no. of years for which data are given.

eg:- If we are given monthly data for 5 years, then we shall first obtain total for each month for five years and divide each total by 5 to obtain an average.

4) Obtain an Average of monthly averages by dividing the total of monthly average by 12.

5) Taking the average of monthly averages as 100, compute the percentage of various monthly averages as follows:

Seasonal Index for January =

$$\frac{\text{Monthly average for January}}{\text{Average of monthly Averages}} \times 100$$

If instead of the average of each month, the totals of each month are obtained, we will get the same result.

Merits & Limitations.

This is the simplest of all methods of measuring Seasonality. Most economic series have trends and, therefore, the seasonal index computed by this method is actually an index of trends and seasonals. Furthermore, the effects of cycles on the original values may or may not be eliminated by the averaging process. This depends on the direction of the cycle.

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and the term of ^{the} average, i.e., on the number of months included in the average. Thus this method is seldom of any value in its simplest form. The method only serves the purpose where no definite trend exists.

Problems

Consumption of monthly electric power in million of kwh for street lighting in a big city during 2013-17 is given below:

Year	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep
2013	318	281	278	250	231	216	223	245	269
2014	342	309	299	268	249	236	242	262	288
2015	367	328	320	287	269	251	259	284	309
2016	392	349	342	311	290	273	282	305	328
2017	420	378	370	334	314	296	305	330	356

Oct Nov Dec

302	325	347
321	342	364
345	367	394
364	389	417
396	422	452

Find out seasonal variation by the method

Happiness is a by product of an effort to make someone else happy